

Yongye(Felix) Hu

📞 510-277-7666 | ✉️ hyy.felix@gmail.com | 🌐 felix-hyy | 👤 felix-hu.space

March 26, 2026

Jeff Mahler
Ambi Robotics
Berkeley, CA

Dear Dr. Mahler,

I am writing to apply for the Senior Robotics Research Engineer role at Ambi Robotics. As a UC Berkeley MEng graduate in Robotics Engineering, I bring direct experience building ROS2-based control systems, integrating industrial perception hardware, and programming multi-axis robotic arms—skills well-suited to extending AmbiOS core libraries.

On the L.I.F.T hybrid flying-driving platform at Berkeley, I developed a ROS2 node graph coupling an MPC controller with a physics simulator to achieve full-trajectory obstacle avoidance with real-time state logging. I also designed a unified drivetrain supporting both aerial and ground locomotion through the same motor-gear system, reducing mechanical complexity while maintaining reliable mode transitions. That tight integration of motion planning, real-time controls, and hardware co-design parallels the architecture Ambi deploys across its sortation fleet through AmbiOS.

My internship at Wuhan Xiaoxian AI Technology grounded that controls background in production constraints. I proposed and built a magnetically-triggered encoder-based imaging system that replaced low-precision pulse triggering, achieving ± 1 mm positional accuracy and cutting point-cloud data loss by 30%. Alongside that sensor work, I wrote embedded C and ladder-logic programs to automate multi-axis motion on a six-axis CRP2 robotic arm for fuel rod testing. Delivering reliable perception and controls on real hardware under real deadlines reinforced the measurement-driven discipline that scales in a fleet environment like Ambi's 150,000+ operating-hour network.

Ambi's trajectory from Dex-Net research to a production-scale Physical AI platform aligns with my own goal of bridging rigorous controls engineering and AI-driven autonomy. I would welcome the opportunity to discuss how my robotics foundation can contribute to your research roadmap.

Sincerely,
Felix Hu